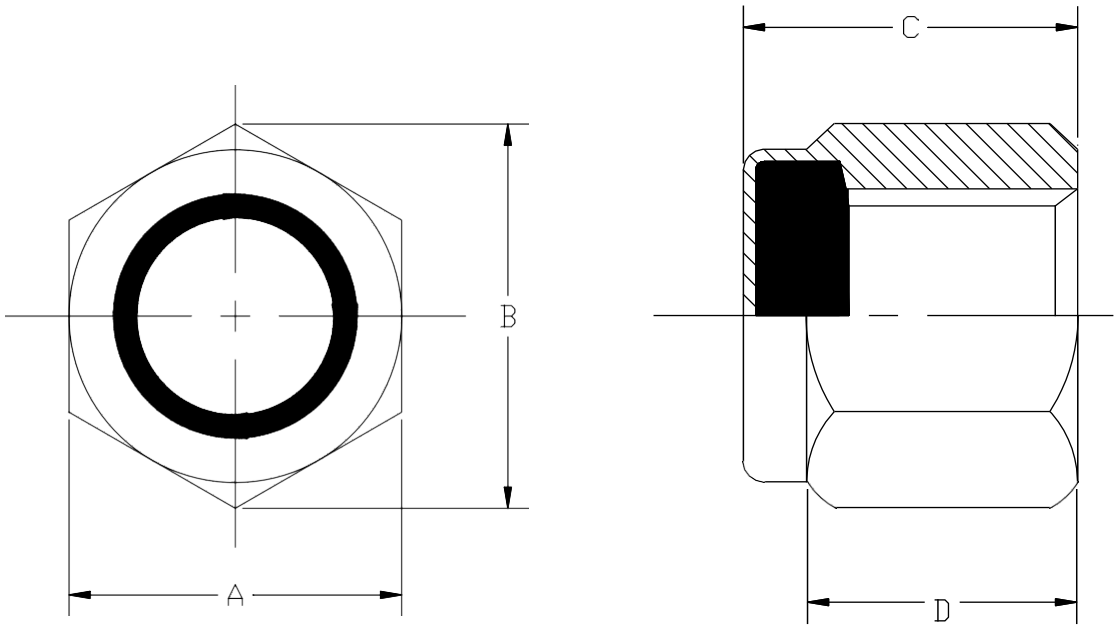
INCH-POUND
MIL-DTL-45913/3C
w/ AMENDMENT 1
21 February 2024
SUPERSEDING
MIL-DTL-45913/3C
20 May 2021

# DETAIL SPECIFICATION SHEET

# NUT, SELF-LOCKING, HEXAGON, NON-METALLIC LOCKING FEATURE, 250 °F, UNC/UNF-3B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-45913.




(All dimensions are in inches.)

FIGURE 1. Nut, self-locking, hex, non-metallic locking feature UNC/UNF-3B.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-IES, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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REQUIREMENTS:

MATERIAL:

Steel and alloy steel shall be in accordance with SAE J995, grade 5 or 8, as specified in part-number.
Corrosion resisting steel (stainless steel) shall be in accordance with ASTM F594, Group 1 or Group 2, have a minimum ultimate tensile strength of 70 KSI and be non-magnetic.
Copper alloy (brass) shall be in accordance with SAE AMS4507, H02 shall have a minimum ultimate tensile strength of 58 KSI and be of a half hard temper.
Aluminum alloy shall be 6061 in accordance with SAE AMS4117 and have a minimum ultimate tensile strength of 42 KSI or as specified in contract.

PLATING/FINISH:

See Table III.

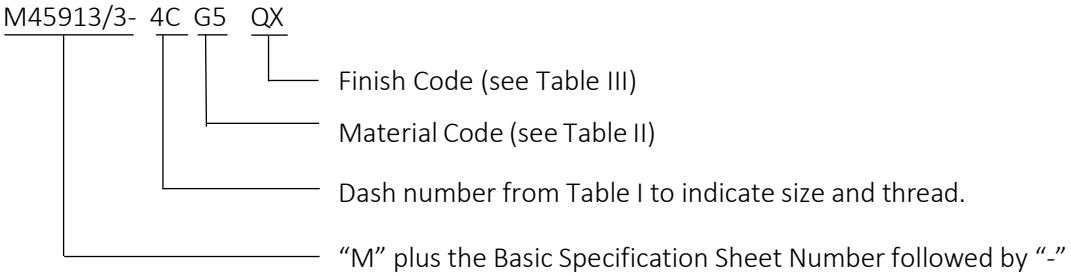
SURFACE TEXTURE. Shall be Ra 125 microinches in accordance with ASME B46.1.

THREADS. Shall be coarse or fine threaded, class 3B, as specified in part-number, in accordance with ASME B1.1.

WORKMANSHIP. Parts shall be manufactured and processed with a level of care and workmanship befitting this type of product intended for general purpose military application. Parts shall be free from dirt, grease, loose or deposited foreign materials, sharp edges, burrs, chips, physical malformations or anything that might prevent the part from reliably performing its intended function.

PART NUMBER. The part number shall consist of the basic number of this specification sheet followed by a dash, a dash number from Table I, a material code from Table II, and a finish/plate code (when required) from Table III.

EXAMPLE OF PART NUMBER:




M45913/3-4CG5QX = Nut Self-Locking Hexagon, non-metallic locking feature, .250-20 UNC-3B thread, Grade 5 steel, Zinc-Nickel, Type K16 plated.



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TABLE I. Dash numbers and dimensions.

| Dash No. | Thread Size | A | | B | C | | D |
|---|---|---|---|---|---|---|---|
| | | Width Across Flats | | Corners | Thickness | | |
| | | MAX | MIN | REF | MAX | MIN | REF |
| 00F | .060-80 UNF-3B | .111 | .104 | .116 | .080 | .055 | .045 |
| 01C | .073-64 UNC-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 01F | .073-72 UNF-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 02C | .086-56 UNC-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 02F | .086-64 UNF-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 03C | .099-48 UNC-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 03F | .099-56 UNF-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 04C | .112-40 UNC-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 04F | .112-48 UNF-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 05C | .125-40 UNC-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 05F | .125-44 UNF-3B | .251 | .243 | .268 | .153 | .133 | .081 |
| 06C | .138-32 UNC-3B | .313 | .305 | .339 | .188 | .168 | .103 |
| 06F | .138-40 UNF-3B | .313 | .305 | .339 | .188 | .168 | .103 |
| 08C | .164-32 UNC-3B | .345 | .336 | .374 | .239 | .219 | .140 |
| 08F | .164-36 UNF-3B | .345 | .336 | .374 | .239 | .219 | .140 |
| 010C | .190-24 UNC-3B | .376 | .367 | .410 | .249 | .229 | .140 |
| 010F | .190-32 UNF-3B | .376 | .367 | .410 | .249 | .229 | .140 |
| 012C | .216-24 UNC-3B | .439 | .430 | .482 | .328 | .298 | .225 |
| 012F | .216-28 UNF-3B | .439 | .430 | .482 | .328 | .298 | .225 |
| 4C | .250-20 UNC-3B | .439 | .428 | .482 | .328 | .298 | .225 |
| 4F | .250-28 UNF-3B | .439 | .428 | .482 | .328 | .298 | .225 |
| 5C | .312-18 UNC-3B | .502 | .489 | .552 | .359 | .329 | .250 |
| 5F | .312-24 UNF-3B | .502 | .489 | .552 | .359 | .329 | .250 |
| 6C | .375-16 UNC-3B | .564 | .551 | .622 | .468 | .438 | .335 |
| 6F | .375-24 UNF-3B | .564 | .551 | .622 | .468 | .438 | .335 |
| 7C | .437-14 UNC-3B | .627 | .616 | .698 | .468 | .438 | .324 |
| 7F | .437-20 UNF-3B | .627 | .616 | .698 | .468 | .438 | .324 |
| 8C | .500-13 UNC-3B | .752 | .736 | .837 | .609 | .579 | .464 |
| 8F | .500-20 UNF-3B | .752 | .736 | .837 | .609 | .579 | .464 |
| 9C | .562-12 UNC-3B | .877 | .861 | .978 | .656 | .626 | .469 |
| 9F | .562-18 UNF-3B | .877 | .861 | .978 | .656 | .626 | .469 |
| 10C | .625-11 UNC-3B | .940 | .922 | 1.051 | .765 | .735 | .593 |
| 10F | .625-18 UNF-3B | .940 | .922 | 1.051 | .765 | .735 | .593 |
| 12C | .750-10 UNC-3B | 1.064 | 1.052 | 1.191 | .890 | .860 | .742 |
| 12F | .750-16 UNF-3B | 1.064 | 1.052 | 1.191 | .890 | .860 | .742 |
| 14C | .875-9UNC-3B | 1.252 | 1.239 | 1.403 | .999 | .969 | .790 |
| 14F | .875-14 UNF-3B | 1.252 | 1.239 | 1.403 | .999 | .969 | .790 |
| 16C | 1.000-8UNC-3B | 1.440 | 1.427 | 1.615 | 1.078 | 1.016 | .825 |
| 16F | 1.000-12 UNF-3B | 1.440 | 1.427 | 1.615 | 1.078 | 1.016 | .825 |
| 18C | 1.125-7UNC-3B | 1.627 | 1.614 | 1.826 | 1.203 | 1.141 | .930 |
| 18F | 1.125-12 UNF-3B | 1.627 | 1.614 | 1.826 | 1.203 | 1.141 | .930 |
| 20C | 1.250-7UNC-3B | 1.815 | 1.801 | 2.038 | 1.422 | 1.360 | 1.125 |
| 20F | 1.250-12 UNF-3B | 1.815 | 1.801 | 2.038 | 1.422 | 1.360 | 1.125 |
| 22C | 1.375-6UNC-3B | 2.008 | 1.973 | 2.232 | 1.609 | 1.547 | 1.282 |
| 22F | 1.375-12 UNF-3B | 2.008 | 1.973 | 2.232 | 1.609 | 1.547 | 1.282 |
| 24C | 1.500-6UNC-3B | 2.197 | 2.159 | 2.444 | 1.640 | 1.578 | 1.313 |
| 24F | 1.500-12 UNF-3B | 2.197 | 2.159 | 2.444 | 1.640 | 1.578 | 1.313 |



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TABLE II. Material coding.

| MATERIAL CODE | |
|---|---|
| Steel Grade 5 | G5 |
| Steel Grade 8 | G8 |
| CRES 303 or 303SE | S3 |
| Alloy Group 1 or 2 from ASTM F594 | S6 |
| Brass | BB |
| Aluminum | AA |

TABLE III. Finish coding.

| Finish | Code | Specification |
|---|---|---|
| Cadmium /1 | C | SAE-AMS-QQ-P-416 Type II, Class 2 or 3 |
| Zinc | Z | ASTM B633 Type II /2, Fe/Zn 8 |
| Zinc (hex chrome-free) | F | ASTM F1941/F1941M Fe/Zn 8CN, sealant optional |
| Black Oxide | B | MIL-DTL-13924, Class 1 or MIL-DTL-495 /3 (for brass) |
| Nickel | N | SAE-AMS-QQ-N-290, Class 2 |
| Zinc-Nickel (Type K16) | QX | MIL-PRF-32647, Type K16 |
| Zinc-Nickel (Type K20) | QW | MIL-PRF-32647, Type K20 /2 |
| Anodized | no code (aluminum only) | MIL-PRF-8625, Type IC or IIB, Class 1 (clear, no color); sealers shall not contain hexavalent chromium compounds |
| Passivate | no code (stainless only) | SAE-AMS 2700, Method 1, Type 6 or 7, or Method 2, Class 1 (for CRES 303 or 303SE, Method 2 only) |
| Uncoated | no code | with dry to touch lubricant (for steel), or uncoated bright finish (for brass), hexavalent chromium prohibited |

1/ Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL- PRF-32647, Type K16 is a suitable replacement with design authority approval.
2/ Type II coating per ASTM B633 contains hexavalent chromium which is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc with hexavalent chromium free passivation per ASTM F1941/F1941M Fe/Zn 8CN, sealant optional or Zinc-Nickel plate per MIL- PRF-32647, Type K20 are suitable replacements with design authority approval.
3/ Hexavalent chromium compounds shall be prohibited.


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NOTES:

1. Dimensions in inches, unless specified, tolerance: ±.016.
2. Interpret drawing in accordance with ASME Y14.5.
3. Break all Sharp edged .003 to .005 and remove all burrs and slivers.
4. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.
5. Unless otherwise specified, issues of referenced documents are those in effect at the time of solicitation.
6. Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - OS
Air Force - 71

Preparing activity:
Army - AT

(Project 5310-2024-008)

Review activities:
Army - AV, CR4, EA, GL, MI
Navy - AS

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